Appendix

To facilitate the discussion of the proof of the multivariate CLT in **Theorem 1** and obtain the exact finite-sample expectation in **Theorem 2**, we reintroduce the required statistics. The logrank statistic at time $t \in [0, \tau]$, denoted by $D(t)$, is expressed as:

$$D(t) = \sum_{i=1}^{n} \Delta_i(t) \left\{ Z_i - \frac{\sum_{j=1}^{n} Z_j Y_j(t, X_i(t))}{\sum_{j=1}^{n} Y_j(t, X_i(t))} \right\} = \sum_{i=1}^{n} \int_0^t N_i(t, dx) \left\{ Z_i - \frac{\sum_{j=1}^{n} Z_j Y_j(t, x)}{\sum_{j=1}^{n} Y_j(t, x)} \right\}$$

where $Y_i(t, x) = I(X_i(t) \geq x)$, and $N_i(t, x) = I(\Delta_i(t) = 1, X_i(t) \leq x)$ for $x \in [0, t]$. We can decompose $D(t)$ into two components by adding and subtracting the appropriate compensator from the counting process:

$$\begin{aligned}
D(t) &= \overline{D}(t) + \widetilde{D}(t) \\
\overline{D}(t) &= \sum_{i=1}^{n} \int_0^t \left(N_i(t, dx) - e^{\theta Z_i} \lambda_0(x) Y_i(t, x)\, dx \right) \left\{ Z_i - \frac{\sum_{j=1}^{n} Z_j Y_j(t, x)}{\sum_{j=1}^{n} Y_j(t, x)} \right\} \\
\widetilde{D}(t) &= \sum_{i=1}^{n} \int_0^t e^{\theta Z_i} \lambda_0(x) Y_i(t, x) \left\{ Z_i - \frac{\sum_{j=1}^{n} Z_j Y_j(t, x)}{\sum_{j=1}^{n} Y_j(t, x)} \right\} dx
\end{aligned}$$

where $e^{\theta Z} = Z e^{\theta} + (1 - Z)$. For $t \in [0, \tau]$ and $x \in [0, t]$, we define the treatment-wise risk set averages $\hat{m}_1(t, x) = n^{-1} \sum_{j=1}^{n} Z_j Y_j(t, x)$, $\hat{m}_0(t, x) = n^{-1} \sum_{j=1}^{n} (1 - Z_j) Y_j(t, x)$, and the combined risk set average $\hat{m}(t, x) = \hat{m}_1(t, x) + \hat{m}_0(t, x)$. We re-express $\overline{D}(t)$ and $\widetilde{D}(t)$ in terms of these risk set averages:

$$\begin{aligned}
\overline{D}(t) &= \sum_{i=1}^{n} \int_0^t \left(N_i(t, dx) - e^{\theta Z_i} \lambda_0(x) Y_i(t, x)\, dx \right) \left\{ Z_i - \frac{\hat{m}_1(t, x)}{\hat{m}(t, x)} \right\} \\
\widetilde{D}(t) &= n\,(e^{\theta} - 1) \int_0^t \lambda_0(x) \frac{\hat{m}_0(t, x)\, \hat{m}_1(t, x)}{\hat{m}(t, x)}\, dx
\end{aligned}$$

Corresponding to the treatment-wise risk set averages $\hat{m}_z(t, x)$; $z \in \{0, 1\}$, we define their asymptotic counterparts $m_1(t, x) = \mathbb{E}[Z_1 Y_1(t, x)]$ and $m_0(t, x) = \mathbb{E}[(1 - Z_1) Y_1(t, x)]$, and $m(t, x) = m_0(t, x) + m_1(t, x)$ corresponding to the combined average $\hat{m}(t, x)$.

In the rest of this appendix we assume that the subject level observations (U_i, V_i, Z_i) are independent for $i = 1, \ldots, n$. In addition to that we assume the cumulative hazard on the

control arm ($Z = 0$) at the scheduled end of the trial τ is finite: $\Lambda_0(\tau) = \int_0^\tau \lambda_0(x)dx < \infty$. In order to prove the required multivariate CLT, we first establish the following Glivenko-Cantelli type lemmas for the risk set averages.

Lemma 1. *For $t \in [0, \tau]$*

$$\sup_{0 \le x \le t} |\hat{m}_1(t, x) - m_1(t, x)| = O_P(n^{-1/2}) \tag{1}$$

Proof. For $t \in [0, \tau]$, we define the stochastic process

$$\xi(x) = Z\,Y(t, x) = Z\,I(X(t) \ge x), \qquad x \in [0, t]$$

where $X(t) = \min\{(t - U)^+, V\}$, and (U, V, Z) has the same distribution as (U_1, V_1, Z_1). For the empirical measure $\mathbb{P}_n$, we can write

$$\hat{m}_1(t, x) = \mathbb{P}_n \xi(x) \qquad \text{and} \qquad m_1(t, x) = P\xi(x)$$

such that

$$n^{1/2}(\hat{m}_1(t, x) - m_1(t, x)) = n^{1/2}(\mathbb{P}_n - P)\xi(x)$$

Since $Z \in \{0, 1\}$ and $Y(t, x) \in \{0, 1\}$ for all $x \in [0, t]$ is non-decreasing, and right-continuous with left hand limits in $x \in [0, t]$, $\{\xi(x) : x \in [0, t]\}$ is a bounded, monotone, càdlàg stochastic process. Furthermore, since $|\xi(x)| \le 1$ for $x \in [0, t]$,

$$P[\,(|\xi(0)| \vee |\xi(t)|)^2\,] \le 1 < \infty$$

By Kosorok Lemma 4.1 ξ is P-Donsker. Therefore, for $t \in [0, \tau]$, the class of functions

$$\mathbb{F}_t = \{f_x : (u, v, z) \mapsto z\,I(\min\{(t - u)^+, v\} \ge x)\,;\, 0 \le x \le t\}$$

is P-Donsker. Consequently, the empirical process

$$G_n(x) = n^{1/2}(\mathbb{P}_n - P)f_x = n^{1/2}(\hat{m}_1(t,x) - m_1(t,x)) \qquad x \in [0,t]$$

converges weakly in $\ell^\infty([0,t])$ to a tight, mean-zero Gaussian Process. We can then use the continuous mapping theorem with

$$g : \ell^\infty[0,t] \to \mathbb{R}, \qquad g(h) = \|h\|_\infty = \sup_{0 \le x \le t} |h(x)|$$

so that

$$\|G_n\|_\infty = \sup_{0 \le x \le t} |n^{1/2}(\hat{m}_1(t,x) - m_1(t,x))|$$

converges in distribution to $\|G\|_\infty$, where G is the tight Gaussian limit. Since $\|G\|_\infty < \infty$ almost surely, it follows that

$$\sup_{0 \le x \le t} |n^{1/2}(\hat{m}_1(t,x) - m_1(t,x))| = O_P(1)$$
$$\implies \sup_{0 \le x \le t} |(\hat{m}_1(t,x) - m_1(t,x))| = O_P(n^{-1/2})$$

□

Corollary 1. *For $t \in [0,\tau]$*

$$\sup_{0 \le x \le t} |\hat{m}_0(t,x) - m_0(t,x)| = O_P(n^{-1/2}) \tag{2}$$

$$\sup_{0 \le x \le t} |\hat{m}(t,x) - m(t,x)| = O_P(n^{-1/2}) \tag{3}$$

We can now begin to prove the multivariate CLT in **Theorem 1**.

Proof of Theorem 1

The proof has the following three main parts:

1. Express $\overline{D}(t)$ and $\widetilde{D}(t)$ for $t \in [0, \tau]$ as the sum of IID terms plus a remainder that converges to 0 in probability with the correct scaling.

2. Establish the multivariate CLT for the scaled and centered logrank statistics.

3. Obtain the expressions for the asymptotic variances and covariances.

Part 1: Decompose $\overline{D}(t)$ and $\widetilde{D}(t)$ as a sum of IID subject-level contributions plus a remainder

At time $t \in [0, \tau]$, by adding and subtracting $m_1(t,x)/m(t,x)$, we can rewrite $\overline{D}(t)$ as:

$$\begin{aligned}
\overline{D}(t) &= \overline{D}_A(t) + \overline{D}_R(t) \\
\overline{D}_A(t) &= \sum_{i=1}^{n} \int_0^t \Big(N_i(t, dx) - e^{\theta Z_i} \lambda_0(x) Y_i(t,x)\, dx \Big) \left(Z_i - \frac{m_1(t,x)}{m(t,x)} \right) \\
\overline{D}_R(t) &= \sum_{i=1}^{n} \int_0^t \Big(N_i(t, dx) - e^{\theta Z_i} \lambda_0(x) Y_i(t,x)\, dx \Big) \left(\frac{m_1(t,x)}{m(t,x)} - \frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} \right)
\end{aligned}$$

where $\overline{D}_A(t)$ is a sum of IID terms, and we shall prove $n^{-1/2}\overline{D}_R(t) = o_P(1)$. Since $\overline{D}_R(t)$ is the sum of n orthogonal, zero-mean, square integrable martingales, using isometry we can write

$$\mathbb{V}(n^{-1/2}\overline{D}_R(t)) = \mathbb{E}[n^{-1/2}\overline{D}_R(t)]^2 = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^{n} \int_0^t \left(\frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)} \right)^2 e^{\theta Z_i} \lambda_0(x) Y_i(t,x) dx \right]$$

We examine the behavior of $\mathbb{E}[n^{-1/2}\overline{D}_R(t)]^2$ depending on the size of the risk set average $m(\hat{t}, x)$, and its asymptotic counterpart $m(t,x)$. On the set $A_n(t) = \{x \in [0,t] : \hat{m}(t,x), m(t,x) > \delta_t\}$ for some $\delta_t > 0$:

$$\begin{aligned}
\left| \frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)} \right| &\leq \left| \frac{1}{\hat{m}(t,x)} \right| |\hat{m}_1(t,x) - m_1(t,x)| + \left| \frac{m_1(t,x)}{m(t,x)} \right| \left| \frac{1}{\hat{m}(t,x)} \right| |\hat{m}(t,x) - m(t,x)| \\
&\leq \frac{1}{\delta_t} \sup_{0 \leq x \leq t} |\hat{m}_1(t,x) - m_1(t,x)| + \frac{1}{\delta_t} \sup_{0 \leq x \leq t} |\hat{m}(t,x) - m(t,x)|
\end{aligned}$$

using $m_1(t,x)/m(t,x) \leq 1$. Since $(a+b)^2 \leq 2(a+b)^2$, we can write

$$\left[\frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)}\right]^2 \leq \frac{2}{\delta_t^2}\left[\Big(\sup_{0\leq x\leq t}|\hat{m}_1(t,x)-m_1(t,x)|\Big)^2 + \Big(\sup_{0\leq x\leq t}|\hat{m}(t,x)-m(t,x)|\Big)^2\right]$$

Using this in the expression for $\mathbb{E}[n^{-1/2}\overline{D}_R(t)]^2$ on the set $A_n(t)$, we can write

$$\begin{aligned}\mathbb{E}[n^{-1/2}\overline{D}_R(t)]^2 &\leq \frac{2}{\delta_t^2}\mathbb{E}\Bigg[\int_0^t \Bigg(\Big(\sup_{0\leq x\leq t}|\hat{m}_1(t,x)-m_1(t,x)|\Big)^2 \\ &\qquad + \Big(\sup_{0\leq x\leq t}|\hat{m}(t,x)-m(t,x)|\Big)^2\Bigg)\lambda_0(x)\left(\frac{1}{n}\sum_{i=1}^n e^{\theta Z_1}\,Y_i(t,x)\right)dx\Bigg] \\ &\leq \frac{2\,(1+e^\theta)}{\delta_t^2}\Lambda_0(t)\mathbb{E}\left[\Big(\sup_{0\leq x\leq t}|\hat{m}_1(t,x)-m_1(t,x)|\Big)^2 + \Big(\sup_{0\leq x\leq t}|\hat{m}(t,x)-m(t,x)|\Big)^2\right]\end{aligned}$$

using $e^{\theta Z_i} \leq (1+e^\theta)$ and $Y_i(t,x) \in \{0,1\}$ for all $x \in [0,t]$. Then using the results (1) and (3), since $0 \leq |\hat{m}_1(t,x)-m_1(t,x)|, |\hat{m}(t,x)-m(t,x)| \leq 1$, we can conclude

$$\mathbb{E}\left[\Big(\sup_{0\leq x\leq t}|\hat{m}_1(t,x)-m_1(t,x)|\Big)^2\right] \to 0\ ,\ \mathbb{E}\left[\Big(\sup_{0\leq x\leq t}|\hat{m}(t,x)-m(t,x)|\Big)^2\right] \to 0$$

Therefore, on the set $A_n(t)$,

$$\mathbb{E}[n^{-1/2}\overline{D}_R(t)]^2 \to 0 \implies n^{-1/2}\overline{D}_R(t) \xrightarrow{P} 0$$

Instead of directly considering the set $A_n^c(t)$, we consider the two sets $B_{n1}(t) = \{x \in [0,t] : m(t,x) < 2\delta_t\}$ and $B_{n2}(t) = \{x \in [0,t] : \hat{m}(t,x) < \delta_t, m(t,x) \geq 2\delta_t\}$. For the second of the two sets:

$$\begin{aligned}P[B_{n2}(t)] &= P\big[\{x \in [0,t] : \hat{m}(t,x) < \delta_t, m(t,x) \geq 2\delta_t\}\big] \\ &\leq P\big[\{x \in [0,t] : |\hat{m}(t,x)-m(t,x)| > \delta_t\}\big] \to 0\end{aligned}$$

using (3). Therefore, on the set $B_{n2}(t)$,

$$\mathbb{E}[n^{-1/2}\overline{D}_R(t)]^2 \leq (1+e^\theta)\Lambda_0(t)P\big[\{x \in [0,t] : |\hat{m}(t,x)-m(t,x)| > \delta_t\}\big] \to 0$$

using

$$\left[\frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)}\right]^2 \leq 1\,, \qquad e^{\theta Z_i} \leq (1+e^{\theta})\,, \qquad Y_i(t,x) \in \{0,1\}$$

On $B_{n1}(t) = \{x \in [0,t] : m(t,x) < 2\delta_t\}$:

$$\begin{aligned}\mathbb{E}[n^{-1/2}\overline{D}_R(t)]^2 &\leq \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n}\int_0^t e^{\theta Z_i}\lambda_0(x)Y_i(t,x)dx\right] \\ &\leq (1+e^{\theta})\,\Lambda_0(t)\,m(t,x) < 2(1+e^{\theta})\,\Lambda_0(t)\,\delta_t\end{aligned}$$

using $\left[\frac{\hat{m}_1(t,x)}{\hat{m}(t,x)} - \frac{m_1(t,x)}{m(t,x)}\right]^2 \leq 1$. Noting that $\delta_t > 0$ can be chosen to be arbitrarily small, on the set $B_{n1}(t)$ we have,

$$\mathbb{E}[\,n^{-1/2}\overline{D}_R(t)]^2 \to 0.$$

Combining these three situations, we have

$$\mathbb{E}[n^{-1/2}\overline{D}_R(t)]^2 \to 0 \implies n^{-1/2}\overline{D}_R(t) = o_P(1)$$

Therefore, for $t \in [0,\tau]$

$$n^{-1/2}\overline{D}(t) = n^{-1/2}\overline{D}_A(t) + o_P(1)$$

We treat $\widetilde{D}(t)$ similarly: for $t \in [0,\tau]$:

$$\begin{aligned}\widetilde{D}(t) &= n\,(e^{\theta}-1)\int_0^t \lambda_0(x)\,\frac{\hat{m}_0(t,x)\,\hat{m}_1(t,x)}{\hat{m}(t,x)}\,dx \\ &= n\,(e^{\theta}-1)\int_0^t \lambda_0(x)\,\phi\Big(\hat{m}_0(t,x)\,,\,\hat{m}_1(t,x)\Big)\,dx\end{aligned}$$

where $\phi(a,b) = ab/(a+b)$. Using a first order asymptotic linear expansion of $\phi\big(\widehat{\mathbf{m}}_{01}(t,x)\big)$ where $\widehat{\mathbf{m}}_{01}(t,x) = \big(\hat{m}_0(t,x)\,,\,\hat{m}_1(t,x)\big)$ around $\mathbf{m}_{01}(t,x) = \big(m_0(t,x)\,,\,m_1(t,x)\big)$, we can

write:

$$\phi(\widehat{\mathbf{m}}_{01}(t,x)) - \phi(\mathbf{m}_{01}(t,x)) - \nabla^T\phi(\mathbf{m}_{01}(t,x))\big(\widehat{\mathbf{m}}_{01}(t,x) - \mathbf{m}_{01}(t,x)\big)$$
$$= \big(\nabla^T\phi(\overline{\mathbf{m}}_{01}(t,x)) - \nabla^T\phi(\mathbf{m}_{01}(t,x))\big)\big(\widehat{\mathbf{m}}_{01}(t,x) - \mathbf{m}_{01}(t,x)\big)$$

where $\overline{\mathbf{m}}_{01}(t,x)$ lies on the line segment joining $\widehat{\mathbf{m}}_{01}(t,x)$ and $\mathbf{m}_{01}(t,x)$. We can then rewrite $\widetilde{D}(t)$ for $t \in [0,\tau]$ as:

$$\widetilde{D}(t) = \widetilde{D}_A(t) + \widetilde{D}_R(t)$$
$$\widetilde{D}_A(t) = n(e^\theta - 1)\int_0^t \lambda_0(x)\Big[\phi(\mathbf{m}_{01}(t,x)) + \nabla^T\phi(\mathbf{m}_{01}(t,x))\big(\widehat{\mathbf{m}}_{01}(t,x) - \mathbf{m}_{01}(t,x)\big)\Big]dx$$
$$\widetilde{D}_R(t) = n(e^\theta - 1)\int_0^t \lambda_0(x)\Big[\big(\nabla^T\phi(\overline{\mathbf{m}}_{01}(t,x)) - \nabla^T\phi(\mathbf{m}_{01}(t,x))\big)\big(\widehat{\mathbf{m}}_{01}(t,x) - \mathbf{m}_{01}(t,x)\big)\Big]dx$$

As before, we shall show $n^{-1/2}\widetilde{D}_R(t) = o_P(1)$ by considering its behavior over the same three sets.

$$n^{-1/2}\widetilde{D}_R(t) = (e^\theta - 1)\int_0^t \lambda_0(x)\Big[\big(\nabla^T\phi(\overline{\mathbf{m}}_{01}(t,x)) - \nabla^T\phi(\mathbf{m}_{01}(t,x))\big)\, n^{1/2}\,\big(\widehat{\mathbf{m}}_{01}(t,x) - \mathbf{m}_{01}(t,x)\big)\Big]dx$$

Noting that

$$\nabla\phi(\overline{\mathbf{m}}_{01}(t,x)) - \nabla\phi(\mathbf{m}_{01}(t,x)) = \left[\left(\frac{\overline{m}_1(t,x)}{\overline{m}(t,x)}\right)^2 - \left(\frac{m_1(t,x)}{m(t,x)}\right)^2, \left(\frac{\overline{m}_0(t,x)}{\overline{m}(t,x)}\right)^2 - \left(\frac{m_0(t,x)}{m(t,x)}\right)^2\right]^T$$

where $\overline{m}(t,x) = \overline{m}_0(t,x) + \overline{m}_1(t,x)$, on the set $A_n(t)$ we can use

$$\left|\left(\frac{\overline{m}_z(t,x)}{\overline{m}(t,x)}\right)^2 - \left(\frac{m_z(t,x)}{m(t,x)}\right)^2\right| = o(1)$$
$$n^{1/2}\big(\hat{m}_z(t,x) - m_z(t,x)\big) \le n^{1/2}\sup_{0\le x\le t}\big|\hat{m}_z(t,x) - m_z(t,x)\big| = O_P(1)$$

for $z \in \{0,1\}$ to show that on $A_n(t)$

$$|n^{-1/2}\widetilde{D}_R(t)| \xrightarrow{P} 0$$

For the set $B_{n2}(t) = \{x \in [0,t] : \hat{m}(t,x) < \delta_t, m(t,x) \geq 2\delta_t\}$, here too we have

$$P\big[B_{n2}(t)\big] \to 0$$

We can treat $n^{-1/2}\widetilde{D}_R(t)$ on the set $B_{n1}(t) = \{x \in [0,t] : m(t,x) < 2\delta_t\}$ in a similar manner as before, using

$$\left| \left(\frac{\overline{m}_z(t,x)}{\overline{m}(t,x)} \right)^2 - \left(\frac{m_z(t,x)}{m(t,x)} \right)^2 \right| \leq 1 \qquad \text{for} \qquad z \in \{0,1\}$$

to show that on $B_{n1}(t)$:

$$\begin{aligned} |n^{-1/2}\widetilde{D}_R(t)| \leq (e^{\theta} - 1) & \\ & \sqrt{n}\Big(\sup_{0 \leq x \leq t} |\hat{m}_1(t,x) - m_0(t,x)| + \sup_{0 \leq x \leq t} |\hat{m}_0(t,x) - m_0(t,x)| \Big) \\ & \int_0^t \lambda_0(x) I(\{x \in [0,t] : m(t,x) < 2\delta_t\}) dx \end{aligned}$$

Since for any $t \in [0,\tau]$ we have $m(t,x) > 0$ for almost every $x \in [0,t]$, for each fixed x such that $m(t,x) > 0$, we have $I(\{m(t,x) < 2\delta_t\}) \to 0$ as $\delta_t \downarrow 0$. Moreover

$$0 \leq \lambda_0(x)\, I(\{m(t,x) < 2\delta_t\}) \leq \lambda_0(x),$$

and $\lambda_0(x)$ is integrable on $[0,t]$. Therefore, by the Dominated Convergence Theorem:

$$\int_0^t \lambda_0(x) I(\{x \in [0,t] : m(t,x) < 2\delta_t\}) dx \to 0 \qquad \text{as} \qquad \delta_t \downarrow 0$$

Combining the three cases we have

$$|n^{-1/2}\widetilde{D}_R(t)| = o_P(1) \qquad \text{for} \qquad t \in [0,\tau]$$

Therefore, for $t \in [0,\tau]$

$$n^{-1/2}\widetilde{D}(t) = n^{-1/2}\widetilde{D}_A(t) + o_P(1)$$

Part 2: Multivariate CLT

Finally, we can write: for $t \in [0, \tau]$

$$\overline{D}_A(t) = \sum_{i=1}^{n} \int_0^t \Big(N_i(t, dx) - e^{\theta Z_i} \lambda_0(x) Y_i(t, x)\, dx \Big) \left(Z_i - \frac{m_1(t,x)}{m(t,x)} \right) = \sum_{i=1}^{n} \psi_{\overline{D},i}(t)$$

where

$$\psi_{\overline{D},i}(t) = \int_0^t \Big(N_i(t, dx) - e^{\theta Z_i} \lambda_0(x) Y_i(t, x) dx \Big) \left(Z_i - \frac{m_1(t,x)}{m(t,x)} \right)$$

is a sum of IID terms, and similarly

$$\begin{aligned} \widetilde{D}_A(t) &= n(e^\theta - 1) \int_0^t \lambda_0(x) \Big[\phi(\mathbf{m}_{01}(t,x)) + \nabla^T \phi(\mathbf{m}_{01}(t,x)) \big(\widehat{\mathbf{m}}_{01}(t,x) - \mathbf{m}_{01}(t,x) \big) \Big] dx \\ &= n\nu(t) + \sum_{i=1}^{n} \psi_{\widetilde{D},i}(t) \end{aligned}$$

where

$$\begin{aligned} \nu(t) &= (e^\theta - 1) \int_0^t \lambda_0(x) \frac{m_0(t,x) m_1(t,x)}{m(t,x)} dx \\ \psi_{\widetilde{D},i}(t) &= (e^\theta - 1) \int_0^t \lambda_0(x) \Bigg[\left(\frac{m_1(t,x)}{m(t,x)} \right)^2 \Big((1 - Z_i) Y_i(t,x) - m_0(t,x) \Big) \\ &\qquad + \left(\frac{m_0(t,x)}{m(t,x)} \right)^2 \Big(Z_i Y_i(t,x) - m_1(t,x) \Big) \Bigg] dx \end{aligned}$$

At the fixed calendar times $0 < t_1 < \cdots < t_K \leq \tau$, we can write for $t_k \in \{t_1, \ldots, t_K\}$:

$$\begin{aligned} n^{-1/2} D(t_k) &= n^{-1/2} D_A(t_k) + o_P(1) \\ \implies n^{-1/2} \Big(D(t_k) - n\, \nu(t_k) \Big) &= n^{-1/2} \sum_{i=1}^{n} \Big(\psi_{\overline{D},i}(t_k) + \psi_{\widetilde{D},i}(t_k) \Big) + o_P(1) \end{aligned}$$

where $D_A(t_k) = \overline{D}_A(t_k) + \widetilde{D}_A(t_k)$. We then have the $K \times 1$ vector

$$n^{-1/2} \begin{pmatrix} D_A(t_1) - n\nu(t_1) \\ \vdots \\ D_A(t_K) - n\nu(t_K) \end{pmatrix} = n^{-1/2} \sum_{i=1}^{n} \begin{pmatrix} \psi_{\overline{D},i}(t_1) + \psi_{\widetilde{D},i}(t_1) \\ \vdots \\ \psi_{\overline{D},i}(t_K) + \psi_{\widetilde{D},i}(t_K) \end{pmatrix} = n^{-1/2} \sum_{i=1}^{n} \mathbf{\Psi}_i$$

where $\mathbf{\Psi}_i = \Big(\psi_{\overline{D},i}(t_1) + \psi_{\widetilde{D},i}(t_1), \ldots, \psi_{\overline{D},i}(t_K) + \psi_{\widetilde{D},i}(t_K)\Big)$ are IID $K \times 1$ vectors. Furthermore, $\mathbb{E}\mathbf{\Psi}_i = \mathbf{0}$ for all i since $\psi_{\overline{D},i}(t_k)$ are mean-zero martingales, and $\mathbb{E}\psi_{\widetilde{D},i}(t_k) = 0$ for all $i = 1, \ldots, n$, $t_k \in \{t_1, \ldots, t_K\}$, since

$$\mathbb{E}\Big[(1 - Z_i)Y_i(t, x) - m_0(t, x)\Big] = 0, \qquad \mathbb{E}\Big[Z_i Y_i(t, x) - m_1(t, x)\Big] = 0$$

Finite second moments are ensured by the assumption $\int_0^\tau \lambda_0(x)dx < \infty$: for $t \in \{t_1, \ldots, t_K\}$

$$\mathbb{E}[\psi_{\overline{D},1}(t_k)]^2 = \int_0^{t_k} \left(Z_1 - \frac{m_1(t_k, x)}{m(t_k, x)}\right)^2 e^{\theta Z_1} \lambda_0(x) Y_1(t_k, x) dx \leq (e^\theta + 1)\Lambda_0(\tau) < \infty$$

using $e^{\theta Z_1} \leq (e^\theta + 1)$, $Y_1(t_k, x) \in \{0, 1\}$, and the square term is bounded above by 1. We can similarly show

$$|\psi_{\widetilde{D},1}(t_k)| \leq |e^\theta - 1|\Lambda_0(\tau) \implies \mathbb{E}[\psi_{\widetilde{D},1}(t_k)]^2 < \infty$$

using

$$|(1 - Z_1)Y_1(t_k, x) - m_0(t_k, x)| \leq 1, \ |Z_1 Y_1(t_k, x) - m_1(t_k, x)| \leq 1, \ \left(\frac{m_z(t_k, x)}{m(t_k, z)}\right)^2 \leq 1 \text{ for } z \in \{0, 1\}$$

By the multivariate CLT for IID random vectors

$$n^{-1/2} \begin{pmatrix} D_A(t_1) - n\nu(t_1) \\ \vdots \\ D_A(t_K) - n\nu(t_K) \end{pmatrix} \xrightarrow{D} \mathcal{N}_K(\mathbf{0}, \Sigma)$$

where $\Sigma = \mathbb{V}(\mathbf{\Psi_1})$ Using Slutsky's Lemma to account for the remainder terms, we can complete the proof:

$$n^{-1/2}\begin{pmatrix} D(t_1) - n\nu(t_1) \\ \vdots \\ D(t_K) - n\nu(t_K) \end{pmatrix} \xrightarrow{D} \mathcal{N}_K(\mathbf{0}, \Sigma)$$

Part 3: Expressions for asymptotic variances and covariances

3.1. Expression for $\mathbb{V}[\psi_{\overline{D},1}(t) + \psi_{\widetilde{D},1}(t)]$; $t \in [0, \tau]$

For $t \in [0, \tau]$, $\mathbb{V}[\psi_{\overline{D},1}(t)] + \psi_{\widetilde{D},1}(t)] = \mathbb{V}[\psi_{\overline{D},1}(t)] + \mathbb{V}[\psi_{\widetilde{D},1}(t)] + 2\mathbb{C}\mathbb{o}\mathbb{v}[\psi_{\overline{D},1}(t)]\,,\, \psi_{\widetilde{D},1}(t)]$. We consider individually each of the three terms.

3.1.1. Expression for $\mathbb{V}[\psi_{\overline{D},1}(t)]$; $t \in [0, \tau]$:

For $t \in [0, \tau]$ we have

$$\mathbb{V}[\psi_{\overline{D},1}(t)] = \mathbb{V}\left[\int_0^t \Big(N_1(t, dx) - e^{\theta Z_1}\lambda_0(x)Y_1(t, x)dx\Big)\left(Z_1 - \frac{m_1(t, x)}{m(t, x)}\right)\right]$$

The concerned stochastic integral being a zero-mean, locally square integrable martingale, usingg isometry we can write

$$\mathbb{V}[\psi_{\overline{D},1}(t)] = \mathbb{E}[\psi_{\overline{D},1}(t)]^2 = \mathbb{E}\left[\int_0^t \left(Z_1 - \frac{m_1(t, x)}{m(t, x)}\right)^2 e^{\theta Z_1}\lambda_0(x)Y_1(t, x)dx\right]$$

Using $e^{\theta Z_1} = Z_1 e^{\theta} + (1 - Z_1)$ we have:

$$\mathbb{V}[\psi_{\overline{D},1}(t)] = \int_0^t \lambda_0(x)\left[e^{\theta} m_1(t, x)\left(\frac{m_0(t, x)}{m(t, x)}\right)^2 + m_0(t, x)\left(\frac{m_1(t, x)}{m(t, x)}\right)^2\right]dx$$

3.1.2. Expression for $\mathbb{V}[\psi_{\widetilde{D},1}(t)]$; $t \in [0, \tau]$:

For $t \in [0, \tau]$ we have

$$\mathbb{V}[\psi_{\widetilde{D},1}(t)] = (e^\theta - 1)^2 \mathbb{E}\Bigg(\int_0^t \lambda_0(x) \Bigg[\left(\frac{m_1(t,x)}{m(t,x)} \right)^2 \Big((1 - Z_1) Y_1(t,x) - m_0(t,x) \Big) + \left(\frac{m_0(t,x)}{m(t,x)} \right)^2 \Big(Z_1 Y_1(t,x) - m_1(t,x) \Big) \Bigg] dx \Bigg)^2$$

since $E[(1 - Z_1)Y_1(t,x) - m_0(t,x)] = 0$ and $\mathbb{E}[Z_1 Y_1(t,x) - m_1(t,x)] = 0$. Let

$$\zeta(t,x) = \left(\frac{m_1(t,x)}{m(t,x)} \right)^2 \Big((1 - Z_1) Y_1(t,x) - m_0(t,x) \Big) + \left(\frac{m_0(t,x)}{m(t,x)} \right)^2 \Big(Z_1 Y_1(t,x) - m_1(t,x) \Big)$$

Then

$$\begin{aligned} \mathbb{V}[\psi_{\widetilde{D},1}(t)] &= (e^\theta - 1)^2 \mathbb{E}\left[\int_0^t \int_0^t \lambda_0(x)\, \lambda_0(y)\, \zeta(t,x)\, \zeta(t,y)\, dx\, dy \right] \\ &= (e^\theta - 1)^2 \int_0^t \int_0^t \lambda_0(x)\, \lambda_0(y)\, \mathbb{E}\Big[\zeta(t,x)\, \zeta(t,y) \Big]\, dx\, dy \\ &= 2\, (e^\theta - 1)^2 \int_0^t \int_0^y \lambda_0(x)\, \lambda_0(y)\, \mathbb{E}\Big[\zeta(t,x)\, \zeta(t,y) \Big]\, dx\, dy \end{aligned}$$

Using $Z_1^2 = Z_1$, $(1 - Z_1)^2 = 1 - Z_1$, $Z_1(1 - Z_1) = 0$ and $Y_1(t,x)Y_1(t,y) = Y_1(t,y)$ since $0 \le x \le y \le t$, we can write

$$\begin{aligned} \mathbb{E}\Big[\zeta(t,x)\, \zeta(t,y) \Big] &= \pi^2(t,x)\pi^2(t,y) \Big(m_0(t,y) - m_0(t,x) m_0(t,y) \Big) \\ &+ (1 - \pi(t,x))^2 (1 - \pi(t,y))^2 \Big(m_1(t,y) - m_1(t,x) m_1(t,y) \Big) \\ &- \pi^2(t,x)\, (1 - \pi(t,y))^2\, m_0(t,x)\, m_1(t,y) \\ &- (1 - \pi(t,x))^2\, \pi^2(t,y)\, m_1(t,x)\, m_0(t,y) \end{aligned}$$

where $\pi(t,x) = m_1(t,x)/m(t,x)$.

3.1.3. Expression for $\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t), \psi_{\widetilde{D},1}(t)]$; $t \in [0, \tau]$:

For $t \in [0, \tau]$, we have

$$
\begin{aligned}
&\mathbb{C}\mathrm{ov}[\,\psi_{\overline{D},1}(t)\ ,\ \psi_{\widetilde{D},1}(t)\,] = \mathbb{E}[\psi_{\overline{D},1}(t)\,\psi_{\widetilde{D},1}(t)] \\
&= (e^{\theta}-1)\mathbb{E}\left[\left(\int_0^t N_1(t,dx)(Z_1-\pi(t,x))\right)\left(\int_0^t \lambda_0(y)\zeta(t,y)dx\right)\right] \\
&\qquad - (e^{\theta}-1)\mathbb{E}\left[\left(\int_0^t e^{\theta Z_1}\lambda_0(x)Y_1(t,x)(Z_1-\pi(t,x))dx\right)\left(\int_0^t \lambda_0(y)\zeta(t,y)dx\right)\right] \\
&= (e^{\theta}-1)(\mathcal{I}_1-\mathcal{I}_2)
\end{aligned}
$$

We consider $\mathcal{I}_2$ first:

$$
\begin{aligned}
\mathcal{I}_2 &= \mathbb{E}\left[\left(\int_0^t e^{\theta Z_1}\lambda_0(x)Y_1(t,x)(Z_1-\pi(t,x))dx\right)\left(\int_0^t \lambda_0(y)\zeta(t,y)dx\right)\right] \\
&= \mathbb{E}\left[\left(\int_0^t e^{\theta}\lambda_0(x)Z_1Y_1(t,x)(Z_1-\pi(t,x))dx\right)\left(\int_0^t \lambda_0(y)(1-\pi(t,y))^2Z_1Y_1(t,y)dy\right)\right] \\
&\quad + \mathbb{E}\left[\left(\int_0^t \lambda_0(x)(1-Z_1)Y_1(t,x)(Z_1-\pi(t,x))dx\right)\left(\int_0^t \lambda_0(y)(\pi(t,y))^2(1-Z_1)Y_1(t,y)dy\right)\right]
\end{aligned}
$$

using $e^{\theta Z_1} = Z_1e^{\theta}+(1-Z_1)$ along with $Z_1(1-Z_1)=0$. Since $Z_1^2 = Z_1$, $(1-Z_1)^2 = (1-Z_1)$, and $Y_1(t,x) = I(X_1(t) \geq x)$, we can write

$$
\begin{aligned}
\mathcal{I}_2 &= e^{\theta}\mathbb{E}\left[Z_1\left(\int_0^{X_1(t)} \lambda_0(x)(1-\pi(t,x))dx\right)\left(\int_0^{X_1(t)} \lambda_0(y)(1-\pi(t,y))^2dy\right)\right] \\
&\qquad + \mathbb{E}\left[(1-Z_1)\left(\int_0^{X_1(t)} \lambda_0(x)(-\pi(t,x))dx\right)\left(\int_0^{X_1(t)} \lambda_0(y)(\pi(t,y))^2dy\right)\right] \\
&= e^{\theta}\,p\,\mathbb{E}[A_1(X_1(t))B_1(X_1(t))|Z_1=1] + \,(1-p)\,\mathbb{E}[A_0(X_1(t))B_0(X_1(t))|Z_1=1]
\end{aligned}
$$

where

$$
A_z(u) = \int_0^u \lambda_0(x)(z-\pi(t,x))dx \qquad \text{and} \qquad B_z(u) = \int_0^u \lambda_0(x)(z-\pi(t,x))^2dx
$$

Therefore,

$$\mathcal{I}_2 = e^{\theta}\, p \int_0^t \Big[A_1^{'}(x)B_1(x) + A_1(x)B_1^{'}(x)\Big] P[X_1(t) \geq x | Z_1 = 1]\, dx$$
$$+ (1-p) \int_0^t \Big[A_0^{'}(x)B_0(x) + A_0(x)B_0^{'}(x)\Big] P[X_1(t) \geq x | Z_1 = 0]\, dx$$

We can similarly show that

$$\mathcal{I}_1 = e^{\theta} p \int_0^t A_1^{'}(x)B_1(x)P[X_1(t) \geq x | Z_1 = 1]\, dx + (1-p) \int_0^t A_0^{'}(x)B_0(x)P[X_1(t) \geq x | Z_1 = 0]\, dx$$

Putting them back together, we get

$$\mathbb{C}\mathrm{ov}[\, \psi_{\overline{D},1}(t)\ ,\ \psi_{\widetilde{D},1}(t)\,] = (e^{\theta} - 1)(\mathcal{I}_1 - \mathcal{I}_2)$$
$$= -(e^{\theta} - 1)\Bigg[e^{\theta} p \int_0^t A_1(x)B_1^{'}(x)P[X_1(t) \geq x | Z_1 = 1]\, dx$$
$$+ (1-p) \int_0^t A_0(x)B_0^{'}(x)P[X_1(t) \geq x | Z_1 = 0]\, dx\Bigg]$$
$$= -(e^{\theta} - 1)\Bigg[e^{\theta} p \int_0^t \left(\int_0^x \lambda_0(y)(1 - \pi(t,y))\, dy \right) \lambda_0(x)(1 - \pi(t,x))^2\, F_{U_1}(t-x)\overline{F}_{V_1|Z_1=1}(x) dx$$
$$+ (1-p) \int_0^t \left(\int_0^x \lambda_0(y)(-\pi(t,y))\, dy \right) \lambda_0(x)(\pi(t,x))^2\, F_{U_1}(t-x)\overline{F}_{V_1|Z_1=0}(x) dx\Bigg]$$

since

$$P[X_1(t) \geq x | Z_1 = z] = P[V_1 \geq x\, , t - U_1 \geq x | Z_1 = z]$$
$$= P[U_1 \leq t - x]P[V_1 \geq x | Z_1 = z]$$
$$= F_{U_1}(t-x)\overline{F}_{V_1|Z_1=1}(x)$$

3.2. Expression for $\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_1)+\psi_{\widetilde{D},1}(t_1)\,,\psi_{\overline{D},1}(t_2)+\psi_{\widetilde{D},1}(t_2)]$; $0\le t_1<t_2\le\tau$

For $0\le t_1<t_2\le\tau$:

$$\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_1)+\psi_{\widetilde{D},1}(t_1)\,,\psi_{\overline{D},1}(t_2)+\psi_{\widetilde{D},1}(t_2)]$$
$$=\mathbb{V}[\psi_{\overline{D},1}(t_1)]+\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_1)\,,\,\psi_{\widetilde{D},1}(t_2)]+\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_2)\,,\,\psi_{\widetilde{D},1}(t_1)]+\mathbb{C}\mathrm{ov}[\psi_{\widetilde{D},1}(t_1)\,,\,\psi_{\widetilde{D},1}(t_2)]$$

3.2.1 Expression for $\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_1)\,,\,\psi_{\widetilde{D},1}(t_2)]$ For $0\le t_1<t_2\le\tau$

$$\begin{aligned}
&\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_1)\,,\,\psi_{\widetilde{D},1}(t_2)]=\mathbb{E}[\psi_{\overline{D},1}(t_1)\,\psi_{\widetilde{D},1}(t_2)]\\
&=(e^{\theta}-1)\mathbb{E}\left[\psi_{\overline{D},1}(t_1)\left(\int_0^{t_2}\lambda_0(x)(1-\pi(t_2,x))^2Z_1Y_1(t_2,x)dx\right)\right]\\
&\qquad+(e^{\theta}-1)\mathbb{E}\left[\psi_{\overline{D},1}(t_1)\left(\int_0^{t_2}\lambda_0(x)(\pi(t_2,x))^2(1-Z_1)Y_1(t_2,x)dy\right)\right]\\
&=(e^{\theta}-1)(\mathcal{I}_1+\mathcal{I}_2)
\end{aligned}$$

Since

$$\begin{aligned}
\pi(t,x)=\frac{m_1(t,x)}{m(t,x)}=\frac{\mathbb{E}[Z_1Y_1(t,x)]}{\mathbb{E}[Y_1(t,x)]}&=\frac{pF_{U_1}(t-x)\overline{F}_{V_1|Z_1=1}(x)}{pF_{U_1}(t-x)\overline{F}_{V_1|Z_1=1}(x)+(1-p)F_{U_1}(t-x)\overline{F}_{V_1|Z_1=0}(x)}\\
&=\frac{p\overline{F}_{V_1|Z_1=1}(x)}{p\overline{F}_{V_1|Z_1=1}(x)+(1-p)\overline{F}_{V_1|Z_1=0}(x)}
\end{aligned}$$

we drop the t in $\pi(t,x)$ and use $\pi(x)$ instead. We first consider $\mathcal{I}_1$:

$$\begin{aligned}
\mathcal{I}_1&=\mathbb{E}\left[\left(\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2Z_1Y_1(t_2,x)dx\right)\psi_{\overline{D},1}(t_1)\right]\\
&=\mathbb{E}\left[\left(\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2Z_1Y_1(t_2,x)dx\right)\left(\int_0^{t_1}\Big(N_1(t_1,dy)-e^{\theta Z_1}\lambda_0(y)Y_1(t_1,y)dy\Big)\Big(Z_1-\pi(y)\Big)\right)\right]\\
&=\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2\,\mathbb{E}\left[Z_1Y_1(t_2,x)\int_0^{t_1}N_1(t_1,dy)\,(1-\pi(y))\right]dx\\
&\qquad-\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2\,\mathbb{E}\left[e^{\theta}Z_1Y_1(t_2,x)\int_0^{t_1}\lambda_0(y)Y_1(t_1,y)(1-\pi(y))dy\right]dx
\end{aligned}$$

where the last equality follows by using $Z_1(1-Z_1)=0$. By conditioning on $Z_1=1$, using

$$Y_1(t_2,x)\,Y_1(t_1,y) = I(U_1 \le (t_2-x)\wedge(t_1-y))\,I(V_1 \ge x\vee y)$$
$$Y_1(t_2,x)N_1(t_1,dy) = 0 \qquad \text{if} \qquad y<x$$

and exchanging the integration and expectation using Fubini, we can write

$$\begin{aligned}
\mathcal{I}_1 &= p\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2\left[\int_x^{t_1}(1-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\,f_{V_1|Z_1=1}(y)\,dy\right]dx\\
&- p\,e^{\theta}\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2\left[\int_0^{t_1}\lambda_0(y)\,(1-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\,\overline{F}_{V_1|Z_1=1}(x\vee y)dy\right]dx\\
&= \mathcal{I}_{11}-\mathcal{I}_{12}
\end{aligned}$$

Writing $f_{V_1|Z_1=1}(y)=e^{\theta}\lambda_0(y)\overline{F}_{V_1|Z_1=1}(y)$ in $\mathcal{I}_{11}$, and splitting up the inner integral in $\mathcal{I}_{12}$, we can write

$$\begin{aligned}
\mathcal{I}_1 &= p\,e^{\theta}\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2\left[\int_x^{t_1}\lambda_0(y)(1-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\,\overline{F}_{V_1|Z_1=1}(y)\,dy\right]dx\\
&- p\,e^{\theta}\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2\left[\int_0^{x}\lambda_0(y)\,(1-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\,\overline{F}_{V_1|Z_1=1}(x)dy\right]dx\\
&- p\,e^{\theta}\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2\left[\int_x^{t_1}\lambda_0(y)\,(1-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\,\overline{F}_{V_1|Z_1=1}(y)dy\right]dx\\
&= -\,p\,e^{\theta}\int_0^{t_2}\lambda_0(x)(1-\pi(x))^2\,\overline{F}_{V_1|Z_1=1}(x)\left[\int_0^{x\wedge t_1}\lambda_0(y)\,(1-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\,dy\right]dx
\end{aligned}$$

Similarly, one can show:

$$\mathcal{I}_2 = -(1-p)\int_0^{t_2}\lambda_0(x)\pi^2(x)\,\overline{F}_{V_1|Z_1=0}(x)dy\left[\int_0^{x\wedge t_1}\lambda_0(y)(-\pi(y))\,F_{U_1}((t_2-x)\wedge(t_1-y))\right]dx$$

3.2.2 Expression for $\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_2)\,,\,\psi_{\widetilde{D},1}(t_1)]$

Using the same steps as in **3.2.1**, one can show for $0\le t_1<t_2\le\tau$

$$\mathbb{C}\mathrm{ov}[\psi_{\overline{D},1}(t_2)\,,\,\psi_{\widetilde{D},1}(t_1)] = (e^{\theta}-1)(\mathcal{I}_1-\mathcal{I}_2)$$

where

$$\mathcal{I}_1 = -p\, e^{\theta} \int_0^{t_1} \lambda_0(x)\,(1-\pi(x))^2\, F_{U_1}(t_1-x)\, \overline{F}_{V_1|Z_1=1}(x) \int_0^{x} \lambda_0(y)\,(1-\pi(y))\, dy\, dx$$
$$\mathcal{I}_2 = -(1-p) \int_0^{t_1} \lambda_0(x)\,(\pi(x))^2\, F_{U_1}(t_1-x)\, \overline{F}_{V_1|Z_1=0}(x) \int_0^{x} \lambda_0(y)\,(-\pi(y))\, dy\, dx$$

3.2.3 Expression for $\mathbb{C}\mathrm{ov}[\psi_{\widetilde{D},1}(t_1)\,,\, \psi_{\widetilde{D},1}(t_2)]$

For $0 \le t_1 < t_2 \le \tau$

$$\begin{aligned}
&\mathbb{C}\mathrm{ov}[\psi_{\widetilde{D},1}(t_1)\,,\, \psi_{\widetilde{D},1}(t_2)] = \mathbb{E}[\psi_{\widetilde{D},1}(t_1)\ \psi_{\widetilde{D},1}(t_2)] \\
&= (e^{\theta}-1)^2 \mathbb{E}\left[\left(\int_0^{t_1} \lambda_0(x)\zeta(t,x)dx\right)\left(\int_0^{t_2} \lambda_0(y)\zeta(t,y)dy\right)\right]
\end{aligned}$$

Now,

$$\begin{aligned}
&\mathbb{E}\left[\left(\int_0^{t_1} \lambda_0(x)\zeta(t,x)dx\right)\left(\int_0^{t_2} \lambda_0(y)\zeta(t,y)dy\right)\right] \\
&= \int_0^{t_1}\int_0^{t_2} \lambda_0(x)\lambda_0(y)\Bigg[(1-\pi(x))^2\,(1-\pi(y))^2 \mathbb{C}\mathrm{ov}\Big(Z_1Y_1(t_1,x)\,,\, Z_1Y_1(t_2,y)\Big) \\
&\qquad + (1-\pi(x))^2\,\pi^2(y)\mathbb{C}\mathrm{ov}\Big(Z_1Y_1(t_1,x)\,,\, (1-Z_1)Y_1(t_2,y)\Big) \\
&\qquad + \pi^2(x)\,(1-\pi(y))^2\mathbb{C}\mathrm{ov}\Big((1-Z_1)Y_1(t_1,x)\,,\, Z_1Y_1(t_2,y)\Big) \\
&\qquad + \pi^2(x)\,\pi^2(y)\mathbb{C}\mathrm{ov}\Big((1-Z_1)Y_1(t_1,x)\,,\, (1-Z_1)Y_1(t_2,y)\Big)\Bigg]\, dy\, dx
\end{aligned}$$

Using $Z_1(1-Z_1)$ we can write

$$\begin{aligned}
&\mathbb{C}\mathrm{ov}[\psi_{\widetilde{D},1}(t_1)\,,\, \psi_{\widetilde{D},1}(t_2)] \\
&= (e^{\theta}-1)^2 \int_0^{t_1}\int_0^{t_2} \lambda_0(x)\lambda_0(y)\Bigg[(1-\pi(x))^2\,(1-\pi(y))^2\Big(\mathbb{E}[Z_1Y_1(t_1,x)Y_1(t_2,x)] - m_1(t_1,x)m_1(t_2,y)\Big) \\
&\qquad + (1-\pi(x))^2\,\pi^2(y)\Big(-m_1(t_1,x)m_0(t_2,y)\Big) + \pi^2(x)(1-\pi(y))^2\Big(-m_0(t_1,x)m_1(t_2,x)\Big) \\
&\qquad + \pi^2(x)\pi^2(y)\Big(\mathbb{E}[(1-Z_1)Y_1(t_1,x)Y_1(t_2,x)] - m_1(t_1,x)m_1(t_2,y)\Big)\Bigg]\, dy\, dx
\end{aligned}$$

where

$$\mathbb{E}[Z_1 Y_1(t_1,x)Y_1(t_2,x)] = p\, F_{U_1}((t_1-x)\wedge(t_2-y))\,\overline{F}_{V_1|Z_1=1}(x\vee y)$$

$$\mathbb{E}[(1-Z_1)Y_1(t_1,x)Y_1(t_2,x)] = (1-p)\, F_{U_1}((t_1-x)\wedge(t_2-y))\,\overline{F}_{V_1|Z_1=0}(x\vee y)$$

Proof of Theorem 2

For $t \in [0,\tau]$ and for fixed number n of subjects entering the trial over the course of the accrual period:

$$\widetilde{D}(t) = (e^{\theta}-1)\sum_{i=1}^{n}\int_0^t \lambda_0(x) Z_i Y_i(t,x)\left(Z_i - \frac{\sum_{j\in R(t,x)} Z_j}{n(t,x)}\right)dx$$

using $e^{\theta Z_i} = Z_i e^{\theta} + (1-Z_i)$ and $n(t,x) = \sum_{i=1}^{n} Y_i(t,x)$. Since the summation over $j \in R(t,x)$ includes the ith subject as well, we can write

$$\widetilde{D}(t) = (e^{\theta}-1)\sum_{i=1}^{n}\int_0^t \lambda_0(x) Z_i Y_i(t,x)\left(1 - \frac{1+\sum_{j\in R(t,x)\setminus\{i\}} Z_j}{1+|R(t,x)\setminus\{i\}|}\right)dx$$

Conditioning on $B_i = \{j = 1,\ldots,n : j \neq i, j \in R(t,x)\}$ along with Z_i, and $\{X_i(t) \geq x\}$:

$$\mathbb{E}\widetilde{D}(t) = (e^{\theta}-1)\sum_{i=1}^{n}\int_0^t \lambda_0(x)\mathbb{E}\left[Z_i Y_i(t,x)\left(1 - \frac{1+\mathbb{E}[\sum_{j\in B_i} Z_j | B_i, Z_i, X_i(t)\geq x]}{1+|B_i|}\right)\right]dx$$

Since Z_j for $j \neq i$ are conditionally distributed as Bernoulli(p), conditionally

$$\sum_{j\in B_i} Z_j \sim \text{binomial}(|B_i|, \pi(t,x)) \qquad \text{where} \qquad \pi(t,x) = P[Z_j = 1 | X_j(t) \geq x]$$

which gives us

$$\mathbb{E}\widetilde{D}(t) = n\,(e^{\theta}-1)\int_0^t \lambda_0(x)\mathbb{E}\left[Z_1 Y_1(t,x)(1-\pi(t,x))\frac{|B_1|}{1+|B_1|}\right]dx$$

Conditioned on $Z_1 = 1$ and $\{X_1(t) \geq x\}$, $|B_1| \sim \text{binomial}(n-1, q(t,x))$ where $q(t,x) = P[X_1(t) \geq x | Z_1 = 1]$. Then

$$\begin{aligned}
\mathbb{E}\left[\frac{1}{1+|B_1|}\middle| Z_1 = 1, X_1(t) \geq x\right] &= \sum_{j=0}^{n-1} \frac{1}{j+1}\binom{n-1}{j} q(t,x)^j (1-q(t,x))^{n-1-j} \\
&= \frac{1}{n\,q(t,x)} \sum_{j=1}^{n} \binom{n}{j} q(t,x)^j (1-q(t,x))^{n-j} \\
&= 1 - \frac{(1-q(t,x)^n)}{n\,q(t,x)}
\end{aligned}$$

Plugging this back into $\mathbb{E}\widetilde{D}(t)$ and using $\mathbb{E}[Z_1 Y_1(t,x)] = p\,q(t,x)$, we have

$$\mathbb{E}\widetilde{D}(t) = n\,p\,(e^{\theta}-1) \int_0^t \lambda_0(x)(1-\pi(t,x))q(t,x)\left[1 - \frac{(1-q(t,x)^n)}{n\,q(t,x)}\right] dx$$

Noting that $\pi(t,x) = P[Z_1 = 1 | X_1(t) \geq x] = m_1(t,x)/m(t,x)$, and $q(t,x) = P[X_1(t) \geq x | Z_1 = 1] = m_1(t,x)/p$, we can show $n^{-1/2}(\mathbb{E}\widetilde{D}(t) - n\nu(t)) = o(1)$ for all $t \in [0,\tau]$.